A SMOOTH MORPHISM OF ALGEBRAIC SPACES IS SYNTOMIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence and the proof is the standard presentation reduction naming the definition and the scheme case; unlike its sibling lemma-smooth-flat the cited reference matches the property being concluded.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The definition of smoothness for morphisms of algebraic spaces. used at proof lines 7505-7508. Produces the presentation with h smooth.
- The source-and-target presentation lemma. Lemma lemma-local-source-target (4313-4340), used at proof lines 7506-7508. Supplies the diagram and the surjective etale vertical arrow.
- The scheme case: a smooth morphism is syntomic. Morphisms, Lemma morphisms-lemma-smooth-syntomic, used at proof lines 7508-7510. Gives syntomicity of h.
- Syntomicity read off from the presentation. used at proof lines 7510-7510. Concludes for X -> Y.

Source reference. Original lemma 03MJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4751–4751. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01VD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2442–2442. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *A smooth morphism of algebraic spaces is syntomic.*

Proof. Let $X \to Y$ be a smooth morphism of algebraic spaces. By definition this means there exists a diagram as in Lemma 03MJ with h smooth and surjective vertical arrow a. By Morphisms, Lemma 01VD h is syntomic. Hence $X \to Y$ is syntomic by definition. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.